Mathematics GCE O Level 2020 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0570 Mathematics 1, June 2020, Ordinary / Intermediate Levels, one and a half hours). Answer ALL 50 questions; all questions carry equal marks. Each question has four suggested answers A, B, C and D; mark only one answer for each question. Calculators are allowed.

Multiple choice questions - choose the one correct answer, A, B, C or D.

1. Simplifying 6.42×0.3 gives

a) 1.826
b) 18.26
c) 1.926
d) 19.26

2. The value of the digit 4 in 2.045 is

a) 0.04
b) 0.4
c) 0.004
d) 40

3. Given that $-3 \leq x$, then the symbol $\leq$ means

a) Strictly less than
b) Less than
c) Equal to
d) less than or equal to

4. Evaluating $\frac{14}{3} \div \frac{7}{2}$ gives

a) $\frac{49}{3}$
b) $\frac{8}{7}$
c) $\frac{8}{3}$
d) $\frac{4}{3}$

5. The number 0.000345 in standard form is

a) 34.5×10^{5}
b) 3.45×10^{-4}
c) 3.45×10^{4}
d) 3.45×10^{-3}

6. The number $500{,}737$ given to three significant figures is

a) 500,000
b) 501
c) 501,000
d) 500,700

7. A computer can print 1800 words in 5 minutes. Working at the same rate the number of words that can be printed in 7 minutes is

a) 2520
b) 1275
c) 1440
d) 720

8. Given that $\frac{5}{8}$ of the children in a school are boys, the ratio of boys to girls is

a) 3:8
b) 5:3
c) 5:8
d) 8:3

9. The distance between two points A and B on a map is 7 cm. Given that AB = 21 km on the ground, the scale on the map is:

a) 1:3
b) 1:3,000
c) 1:30,000
d) 1:300,000

10. An infinite set in the following is:

a) $\{1, 3, 5, 7\}$
b) $\{2, 4, 6, 8\}$
c) $\{1, 2, 3, \dots\}$
d) $\{1, 2, 3, \dots, 10\}$

11. Given the set $B = \{1, 2, 3\}$, the number of subsets of B is:

a) 2^4
b) 2^3
c) 2^2
d) 2^1

12. Given the statements: p: He is short. q: He is fat. Then $\sim p \wedge q$ is:

a) He is tall and fat
b) He is short and fat
c) He is tall and thin
d) He is short and thin

13. In the truth table in figure 1, X represented symbolically is:

p	q	X
T	T	T
T	F	T
F	T	T
F	F	F

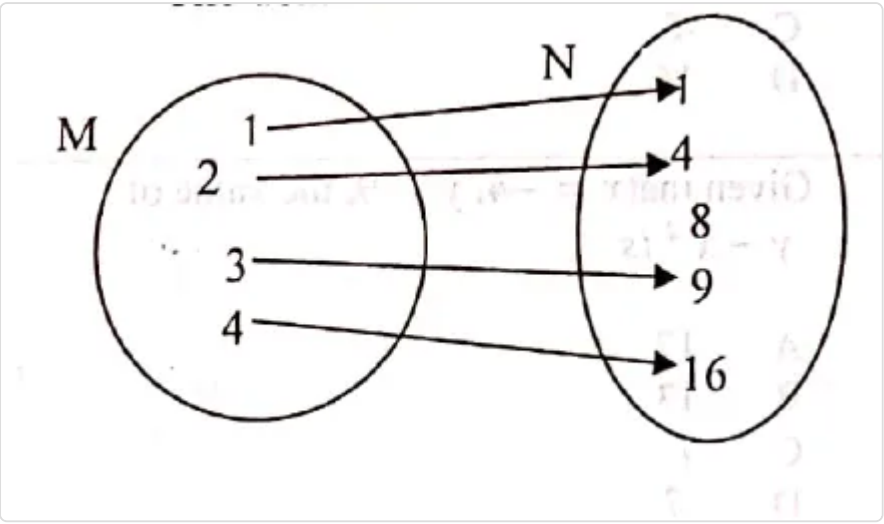
Figure 1

a) $p \vee q$
b) $p \wedge q$
c) $p \Rightarrow q$
d) $p \Leftrightarrow q$

14. Given that $f(x) = x^2 - 5x - 6$, then $f(2) =$

a) 20
b) 12
c) 0
d) -12

15. The arrow diagram in figure 2 shows the relationship between M and N. The domain is

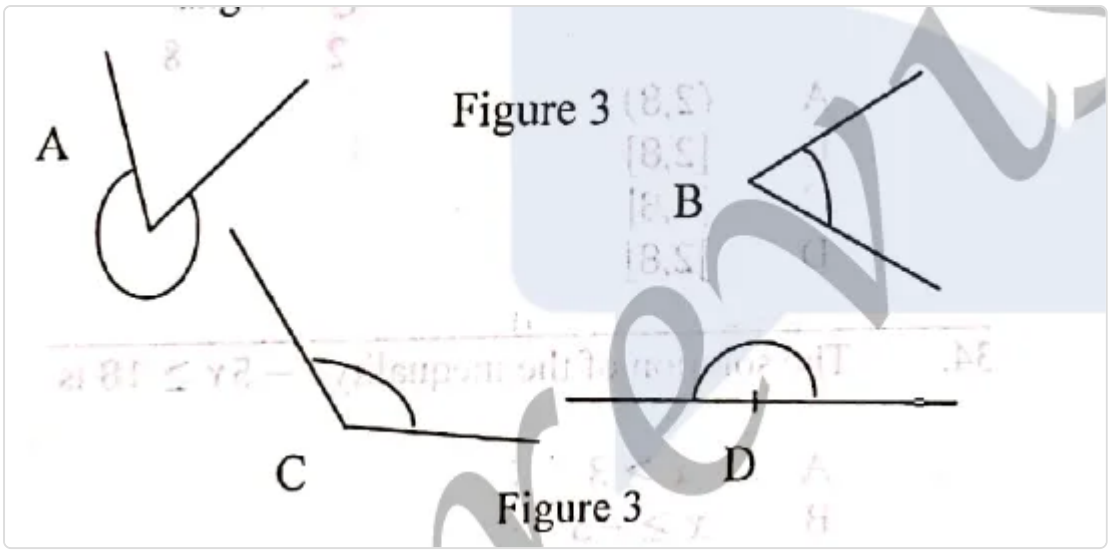


a) $\{1, 4, 9, 8, 16\}$
b) $\{1, 2, 3, 4\}$
c) $\{1, 4, 9, 16\}$
d) $\{1, 4, 3, 8, 16\}$

16. Given the function $f : x \longmapsto 2x - 3$, the inverse function is

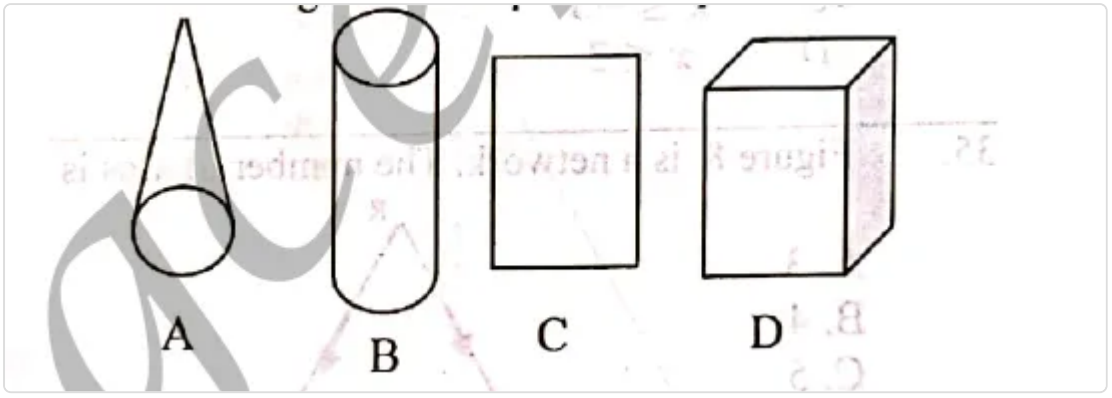
a) $\frac{x-3}{2}$
b) $\frac{x-2}{2}$
c) $\frac{x+3}{2}$
d) $\frac{3x}{2}$

17. In figure 3, the diagram that represents a reflex angle is:



a) A
b) B
c) C
d) D

18. The diagram that represents a plane figure is:

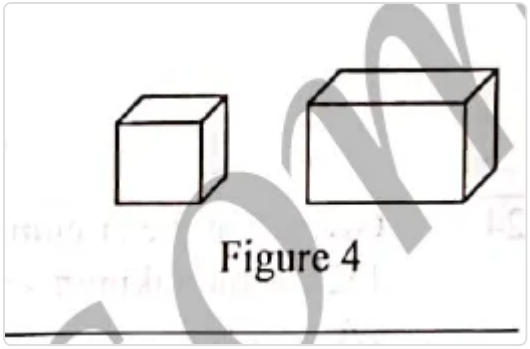


a) A
b) B
c) C
d) D

19. The sum of the interior angles of a triangle is:

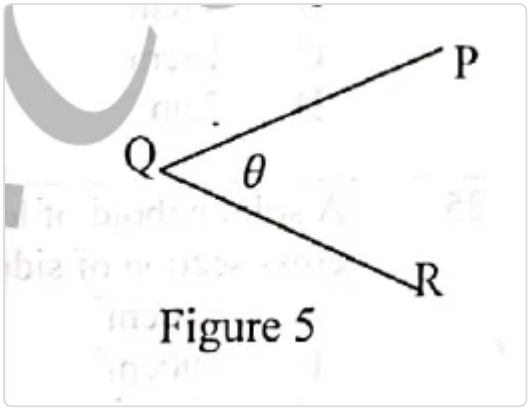
a) 90°
b) 180°
c) 120°
d) 360°

20. Figure 4 shows two similar cubes. The ratio of their lengths is $1:2$. The length of the smaller cube is 5 cm. The volume of the bigger cube is:



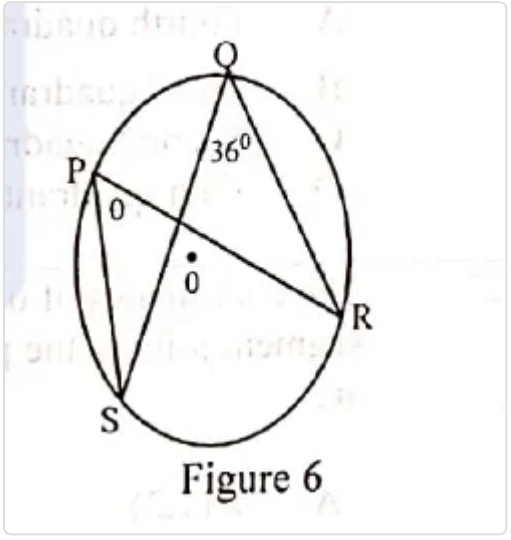
a) $27\ \text{cm}^3$
b) $125\ \text{cm}^3$
c) $100\ \text{cm}^3$
d) $1000\ \text{cm}^3$

21. In figure 5, the lines that enclose angle θ are



a) $P\theta$ and $Q\theta$
b) PR and QR
c) PQ and PR
d) PQ and RQ

22. In figure 6, O is the centre of the circle. Angle $SQR = 36°$. The value of angle θ is:



a) $72°$
b) $36°$
c) $18°$
d) $54°$

23. In figure 7, the value of y is

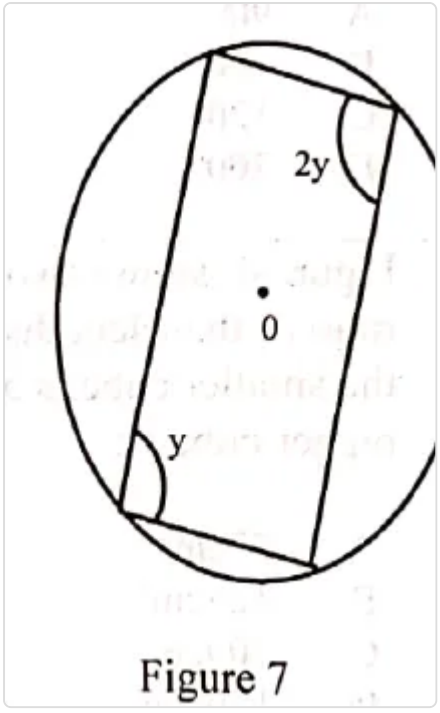


Figure 7

a) 105°
b) 75°
c) 60°
d) 45°

24. Given that the circumference of a circle is 132 cm and taking $\pi = \frac{22}{7}$, its radius is

a) 21 cm
b) 49 cm
c) 14 cm
d) 7 cm

25. A solid cuboid of length 10 cm has a square cross section of side 8 cm. The volume is

a) $640\ \text{cm}^2$
b) $100\ \text{cm}^3$
c) $80\ \text{cm}^2$
d) $640\ \text{cm}^3$

26. The point $P(3, -2)$ in the Cartesian plane $x - y$ is found in the

a) Fourth quadrant
b) Third quadrant
c) Second quadrant
d) First quadrant

27. The coordinates of the midpoint of the line segment joining the points $A(5, 6)$ and $B(-1, -4)$ are

a) A(1,2)
b) A(2,1)
c) A(4,2)
d) A(-2,-1)

28. Given the equations of the straight lines $x + y = 4$ and $y = 1 + x$. The coordinates of the point of intersection are:

a) $(1, 2)$
b) $\left(1\frac{1}{2}, 2\right)$
c) $\left(1\frac{1}{2}, 2\frac{1}{2}\right)$
d) $\left(2\frac{1}{2}, 1\frac{1}{2}\right)$

29. In the expression $2x(5 - 3x)$ the coefficient of x is

a) 10
b) 4
c) -6
d) 16

30. Given that $x = -4$, $y = 9$, the value of $y - x^2$ is

a) 17
b) 13
c) 7
d) -7

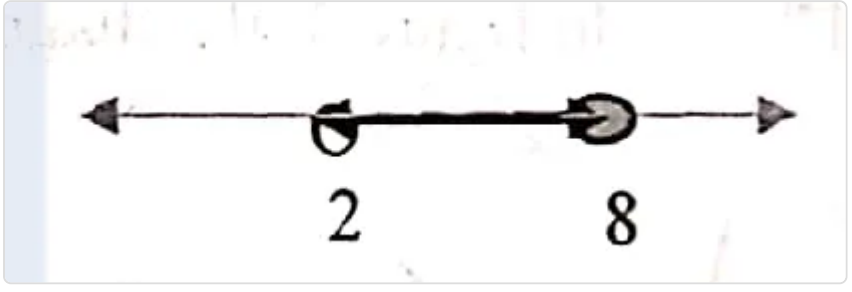
31. Given the expressions $2xy$ and $6x^2y$, the common factor is:

a) $2y^2$
b) $2x^2$
c) $2xy$
d) $2x^2y$

32. Given that $V = \pi r^2 l$, expressing r in terms of V, π and l gives

a) $\frac{\sqrt{V}}{\pi l}$
b) $\sqrt{\frac{V}{\pi l}}$
c) $\left(\frac{V}{\pi l}\right)^2$
d) $\frac{V}{2\pi l}$

33. The interval represented on the number line is



a) $(2, 8)$
b) $[2, 8]$
c) $[2, 8[$
d) $]2, 8]$

34. The solution of the inequality $3 - 5x \geq 18$ is

a) $x \geq 3$
b) $x \geq -3$
c) $x \leq -3$
d) $x \leq 3$

35. Figure 8 is a network. The number of arcs is

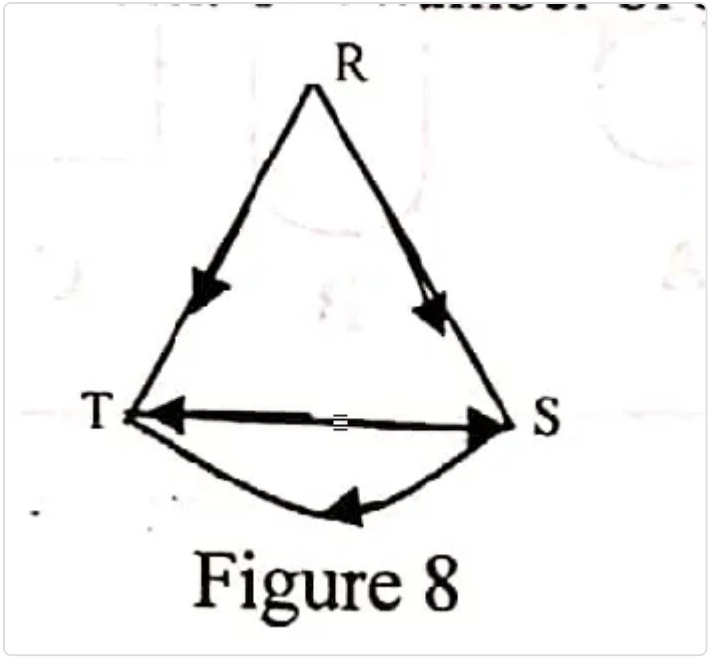


Figure 8

a) 3
b) 4
c) 5
d) 2

36. Given that $5^{2x} = 125$, the value of x is:

a) $\frac{3}{2}$
b) $\frac{2}{3}$
c) 2
d) 3

37. Figure 9 is a right-angled triangle. The value of h is

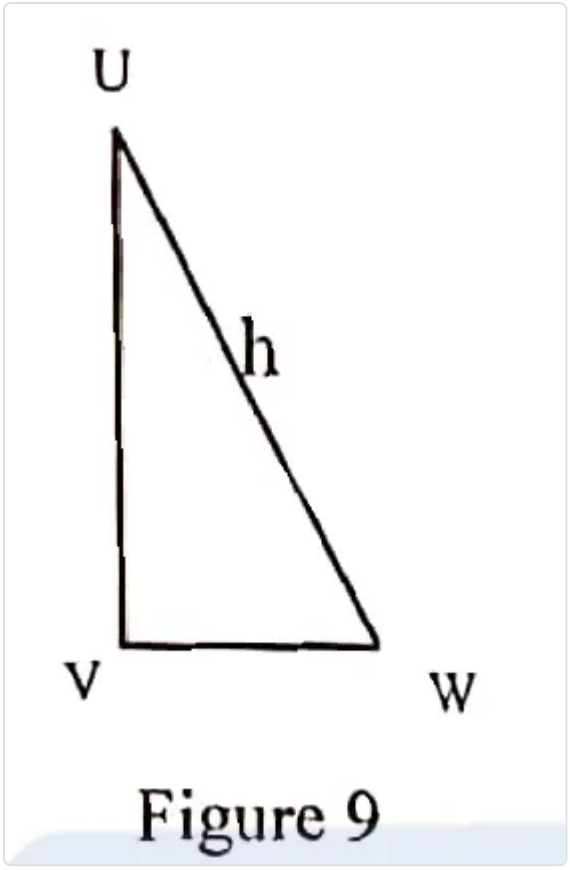


Figure 9

a) 169 cm
b) 13 cm
c) 17 cm
d) 84.5 cm

38. Given that the triangle in figure 10 is right-angled at R, $\sin x$ is:

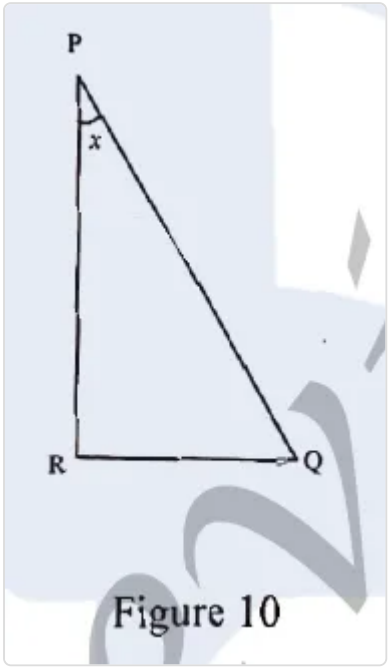
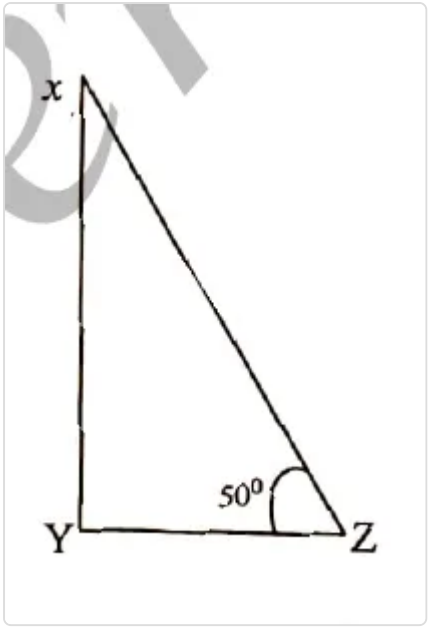


Figure 10

a) $\frac{PR}{RQ}$
b) $\frac{PR}{PQ}$
c) $\frac{RQ}{PQ}$
d) $\frac{RQ}{PR}$

39. Figure 11 shows the angle of elevation of X from Z to be 50°. The angle of depression of Z from X is



a) 50°
b) 40°
c) 90°
d) 140°

40. Given that $\vec{a} = 5\vec{i} + 7\vec{j}$ and $\vec{b} = 4\vec{i} - 5\vec{j}$, then $\vec{a} - 2\vec{b}$ is

a) $-3\vec{i} - 3\vec{j}$
b) $3\vec{i} - 17\vec{j}$
c) $3\vec{i} + 3\vec{j}$
d) $-3\vec{i} + 17\vec{j}$

41. Given the vector $\overrightarrow{Op} = 2\vec{i} + \vec{j}$, the direction of the vector is:

a) 63.4°
b) 26.6°
c) 90°
d) 116.6°

42. Two vectors $a\vec{i} + 2\vec{j}$ and $6\vec{i} + 4\vec{j}$ are said to meet at right angles. The value of a is:

a) $\frac{4}{3}$
b) $\frac{3}{4}$
c) $-\frac{4}{3}$
d) $-\frac{2}{4}$

43. The order of the matrix $m = \begin{pmatrix} 3 & 1 & 0 \\ -5 & 7 & 6 \end{pmatrix}$ is

a) 3×2
b) 2×6
c) 3×6
d) 2×3

44. Given the matrices $M = \begin{pmatrix} 3 & 4 \\ 1 & -2 \end{pmatrix}$ and $N = \begin{pmatrix} 5 & 2 \\ -3 & 1 \end{pmatrix}$, then $M + N =$

a) $\begin{pmatrix} 8 & 6 \\ 4 & 3 \end{pmatrix}$
b) $\begin{pmatrix} 8 & 6 \\ -2 & -2 \end{pmatrix}$
c) $\begin{pmatrix} 8 & 6 \\ -2 & -1 \end{pmatrix}$
d) $\begin{pmatrix} 8 & 6 \\ -3 & -3 \end{pmatrix}$

45. The image of the matrix (x, y), translated by the matrix $(6, 2)$ is $(-3, -2)$. The matrix (x, y) is

a) $(-3, 0)$
b) $(-9, 0)$
c) $(-3, -4)$
d) $(-9, -4)$

46. The mode of the scores $3, 2, 6, 5, 1, 4, 2, 3, 6, 3, 2, 2, 7, 4, 2$ is

a) 1
b) 2
c) 3
d) 5

47. | x | 1-5 | 6-10 | 11-15 | 16-20 | 21-25 |
|---|---|---|---|---|---|
| f | 4 | 1 | 2 | 3 | 0 |

The number of classes in the above frequency distribution is

a) 3
b) 4
c) 5
d) 10

48. The lower quartile of a cumulative frequency curve is at the

a) 25th percentile
b) 50th percentile
c) 75th percentile
d) 100th percentile

49. Given the set $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, the probability of selecting a prime number from S is

a) $\frac{3}{5}$
b) $\frac{3}{10}$
c) $\frac{1}{2}$
d) $\frac{2}{5}$

50. Given the events E and F such that $P(E \cap F) = P(E)P(F)$, then the events E and F are

a) Mutually exclusive
b) Independent
c) Complementary
d) Exhaustive

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